

HARSHIT TIWARI

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RESEARCH INTERESTS

Turbulent convection, atmospheric and astrophysical flows, compressible flows, High-Performance Computing (HPC), quantum turbulence, turbulence and nonlinear dynamics, etc

EDUCATION

-  **Indian Institute of Technology Kanpur** Kanpur, India
Doctor of Philosophy in Physics *August 2021 - February 2026 (Thesis submitted)*
 - Courses on Tapestry of Field Theory, Physics of Turbulence, High-Performance Computing and Advanced Statistical Physics.
 - Current Cumulative Performance Index: 9.56/10
-  **Indian Institute of Technology Kanpur** Kanpur, India
Master of Science in Physics *July 2019 - July 2021*
 - Courses on High Energy Astrophysics, Nuclear and Particle Physics and Quantum Field Theory.
 - Cumulative Performance Index: 8.30/10
-  **Kumaun University** Nainital, India
Bachelor of Science *July 2016 - June 2019*
 - Subjects: Physics, Mathematics, Chemistry
 - Percentage: 66.4%, First Class

RESEARCH EXPERIENCE

-  **Graduate Researcher, Department of Physics, IIT Kanpur** 2021 - Present
Supervisor: Prof. Mahendra Verma, Department of Physics, IIT Kanpur
Co-supervisor: Prof. Rajesh Ranjan, Department of Aerospace Engineering, IIT Kanpur

Thesis Title: *Compressible turbulent convection at extreme Rayleigh numbers*
 - Simulated turbulent compressible convection at extreme Rayleigh numbers, revealing classical heat transport scaling laws.
 - Developed a scalable Python PDE solver with GPU and MPI support for high-performance simulations.
 - Applied advanced numerical methods to study shocks, turbulence, and compressible flows in astrophysics and atmosphere.
-  **Master's Student, Department of Physics, IIT Kanpur** 2020 - 2021
Supervisor: Prof. Pankaj Jain, Department of Physics, IIT Kanpur
Co-supervisor: Prof. J.S. Yadav, Department of Physics, IIT Kanpur

Project: *Theoretical Modelling of Accretion Disk Oscillations*
 - Studied acoustic normal modes in thin accretion disks using an effective Kerr potential.
 - Derived a dispersion relation valid over a range of black hole spin parameters.

PUBLICATIONS

1. **H. Tiwari**, L. Sharma, and M. K. Verma, On the absence of the ultimate regime in turbulent thermal convection, *The Proceedings of the National Academy of Sciences*. 122 (44) e2513474122 (2025).
2. **H. Tiwari**, L. Sharma, and M. K. Verma, Compressible turbulent convection at very high Rayleigh numbers, *International Journal of Heat and Mass Transfer*. 242, 126821 (2025).
3. D. Singh, **H. Tiwari**, L. Sharma, and M. K. Verma, Mathematical formulation of mode-to-mode energy transfers and energy fluxes in compressible turbulence, *Physical Review Fluids*. 10, 114609 (2025).
4. L. Sharma, M. Pathak, **H. Tiwari**, and M. K. Verma, Effect of Prandtl number on turbulent compressible convection, *Physical Review Fluids*. 10, 114611 (2025).
5. L. Sharma, M. Pathak, **H. Tiwari**, and M. K. Verma, Variation of convective heat flux imbalance with Prandtl number, *Center for Turbulence Research Annual Reports Briefs* 2025.
6. D. Singh, **H. Tiwari**, L. Sharma, and M. K. Verma, Scale-by-Scale Energy Transfers and Fluxes in Compressible Turbulence, *Europhysics Letters* (2025). (Under review)
7. **H. Tiwari**, D. Singh, M. K. Verma, and R. Ranjan, Scaling in supersonic turbulence: Energy spectra and fluxes using high-fidelity direct numerical simulations, *Journal of Fluid Mechanics* (2025). (Under review)
8. **H. Tiwari**, N. Kumar, M. K. Verma, and S. Ravichandran, Direct numerical simulations of moist turbulent compressible convection. (Under preparation)
9. **H. Tiwari**, A. Pandey, K. R. Sreenivasan, Scaling laws in four dimensional turbulent convection. (Under preparation)
10. N. Kumar, **H. Tiwari**, M. K. Verma, and S. Ravichandran, Heat transport scaling in Rayleigh-Bénard convection. (Under preparation)
11. A. Pandey, **H. Tiwari**, K. R. Sreenivasan, Transients depend on the dimensionality of thermal convection. (Under preparation)

CONFERENCES AND WORKSHOPS

- Participated in the **Indo-Japanese Workshop for CFD-based AI and HPC 2025** at IIT Kanpur.
- Talk on “Compressible turbulent convection at very high Rayleigh numbers” at the **APS Division of Fluid Dynamics Annual Meeting 2025** in *Houston, Texas, USA*.
- Poster presentation titled “Compressible turbulent convection at extreme Rayleigh numbers” at **The Variable Sun: Past, Present, and Future Perspectives 2025** at *Thiruvananthapuram, India*.
- Talk on “Classical 1/3 Nusselt Scaling in Compressible Convection” at the **1st European Fluid Dynamics Conference (EFDC1) 2024** at *Aachen, Germany*.
- Talk on “Classical 1/3 Nusselt Scaling in Compressible Convection at Extreme Ra” at **ICTS Program on Theoretical and Practical Perspectives in Geophysical Fluid Dynamics 2024**.
- Poster presentation titled “Classical Nusselt 1/3 scaling up to $Ra = 10^{16}$ in turbulent compressible convection” at **HPC Symposium 2024**, *IIT Kanpur*.
- Oral presentation on compressible turbulent convection at the **Research Scholar Day 2024**, organised by the *Department of Physics, IIT Kanpur*.
- Participated in **Frontier Hackathon, March 2024**. We scaled the compressible finite-difference solver *Dhara* on Frontier up to 8192 AMD MI250X GPUs.
- Attended the **ICTS Program on Field Theory and Turbulence 2023**.
- Attended the **ICTS Program on Turbulence: Problems at the interface of Mathematics and Physics 2023**.
- Participated in **NSM GPU Hackathon 2022**. We ported the Quantum Solver (Gross–Pitaevskii equation) sequential code to run on multiple GPUs and scaled it on 64 Nvidia A100 GPUs.
- Attended the **ICTS Summer School on Gravitational-Wave Astronomy 2021**.

TEACHING EXPERIENCE

I have assisted in the following courses at the Indian Institute of Technology Kanpur:

- **PHY441A: Electronics** from August 2021 to May 2022.
- **PHY473A: Computational Physics** from August 2022 to November 2022.
- **PHY113A: Classical Electrodynamics** from March 2023 to November 2023.
- **NPTEL: Scientific Computing using Python** from June 2023 to November 2023.
- **NPTEL: Tapestry of Field Theory: Classical Quantum, Equilibrium, Nonequilibrium Perspectives** from January 2024 to January 2025.
- **PHY461A/462A: Experimental Physics I/II** from January 2024 to May 2025.
- **PHY111A: Undergraduate Lab** from August 2025 to August 2026.

COMPUTATIONAL SKILLS

- **Advanced:** Python, parallel programming, Numba, Paraview
- **Intermediate:** Matlab, Mathematica, CUDA
- **Basic:** C++, Julia, Fortran, R

GRANTS AND AWARDS

- Session chair for “Convection and Buoyancy-Driven Flows: Heat Transfer, Instabilities & Turbulence” at **APS Division of Fluid Dynamics Annual Meeting 2025** in *Houston, Texas, USA*.
- Recipient of the **2025 Division of Fluid Dynamics Enabling Award**, American Physical Society, supporting attendance at the 2025 DFD Annual Meeting, Nov 2025.
- Secured an **All India Rank 74** among 15,000 applicants in the **IIT Joint Admission Test (JAM) 2019**, for admission to the M.Sc. program at the Indian Institute of Technology Kanpur.
- Recipient of **Merit cum Means Scholarship** at Indian Institute of Technology Kanpur, Aug 2019 - May 2021.
- Secured **All India Rank 307** in **Joint Entrance Screening Test (JEST) 2019**.
- Qualified **National Defence Academy (NDA)** entrance exam in 2016.

OTHER EXPERIENCES

1. Coordinator, Adventure Sports Club, IIT Kanpur (2024–25)

- Led club activities, organizing fitness programs, treks, runs, and yoga, boosting participation and visibility.

2. Secretary, Adventure Sports Club, IIT Kanpur (2022–23, 2023–24)

- Assisted in organising events and managing logistics for outdoor activities and training sessions.

3. Election Officer, Hall 7, HEC Elections 2022

- Managed smooth conduct of hostel elections with a voter turnout of ~85%.

REFERENCES

1. **Prof. Mahendra K. Verma**, Department of Physics, IIT Kanpur, India.
 mkv@iitk.ac.in
2. **Prof. Rajesh Ranjan**, Department of Aerospace Engineering, IIT Kanpur, India
 rajeshr@iitk.ac.in
3. **Prof. Shashwat Bhattacharya**, School of Mechanical and Materials Engineering, IIT Mandi, India
 shashwat@iitmandi.ac.in